

Stage 3 Final Report

Business Analyst Contribution

Retention Radar - Customer Churn Analysis
Advanced Telecommunications Solutions (ATS)

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Purpose of this report

This document provides the Business Analyst contribution for the Stage 3 final submission. It translates the Data Analyst's ANN model outputs and the Data Engineer's QA validation results into business findings, churn and retention factors, targeted recommendations, limitations, and video presentation content.

1. Executive Summary

Stage 3 focused on refining and evaluating an Artificial Neural Network (ANN) model to predict customer churn. From a Business Analyst perspective, the main objective is to convert the model results into business insight that can support targeted retention planning and reduce wasted retention effort.

The refined ANN model achieved 73.39% accuracy, 73.46% recall, 59.44% F1-score, 0.8172 ROC-AUC, and 0.5922 PR-AUC at the selected threshold of 0.52. The strongest business value of the model is its ability to rank customers by churn risk, especially because customer churn is an imbalanced problem where accuracy alone can be misleading.

The key churn drivers were MonthlyCharges, tenure, and EstimatedTotalCharges. The strongest action insight is that the riskiest 20% of customers contain 44.7% of all actual churners, which means targeted retention campaigns are more efficient than contacting all customers equally. The most important customer segment for action is short-tenure, high-charge customers, with an actual churn rate of 48.9%.

The Data Engineer QA validation supports confidence in the model outputs. The data hand-off had no missing values, no duplicate rows, no train/test overlap, correct scaling, and matching row counts. Therefore, the BA recommendation is to use the model as a decision-support tool for prioritising retention actions, while also considering customer service history, commercial judgement, and future data improvements.

2. Business Analyst Role and Scope

- Translate technical ANN model results into plain business language for non-technical stakeholders.
- Summarise model performance in terms of business usefulness, not only technical accuracy.
- Identify the main churn and retention factors supported by the model results.
- Recommend practical retention strategies for high-risk customer groups.
- Document limitations and propose future improvements.
- Support the final report and video demonstration with stakeholder-ready explanations.

3. Data and QA Validation Summary

Before interpreting the model results, the data hand-off was checked by the Data Engineer. This is important because model outputs can only be trusted if the input data is clean, consistent, and correctly prepared.

QA check	Result	Business Analyst meaning
Row counts	6,741 total records; 5,392 training; 1,349 testing	The split matches the documented Stage 2 hand-off.
Churn rate balance	26.57% full data; 26.58% training; 26.54% testing	Training and testing sets are representative and comparable.
Train/test overlap	0 overlapping rows	The test results are credible because test customers were not seen during training.
Scaling	Scaling was fit on training data and applied correctly to testing data	No test-data leakage was introduced through scaling.
Missing values	0 blank values across files	No missing-data issue was found at hand-off.
Duplicate records	0 exact duplicate customer rows	No duplicate customers inflated the model results.
Feature columns	Expected feature groups were present and named correctly	The model used the intended Stage 2 feature set.

BA interpretation: the QA results support the reliability of the Stage 3 model outputs. No further data cleaning action is required before using the results in the final report.

4. ANN Model Performance and Business Meaning

The final model was tested on 1,349 held-out customers that were not used for model training or threshold selection. The selected decision threshold was 0.52, chosen to balance precision, recall, and practical retention contact volume.

Metric	Stage 3 result at threshold 0.52	Business interpretation
Accuracy	0.7339 / 73.39%	The model correctly classifies a strong proportion of total customer outcomes.
Precision	0.4991 / 49.91%	Around half of predicted churners are actual churners, so campaigns should still be targeted carefully.
Recall	0.7346 / 73.46%	The model catches roughly three out of four customers who actually churn, which is valuable for retention.
F1-score	0.5944 / 59.44%	This balances precision and recall and shows useful churn-class performance.
ROC-AUC	0.8172	The model is strong at ranking risky customers above safer customers.
PR-AUC	0.5922	Useful for assessing performance on a minority churn class.
Decision threshold	0.52	A business-adjustable cut-off that can be changed based on campaign budget and contact tolerance.

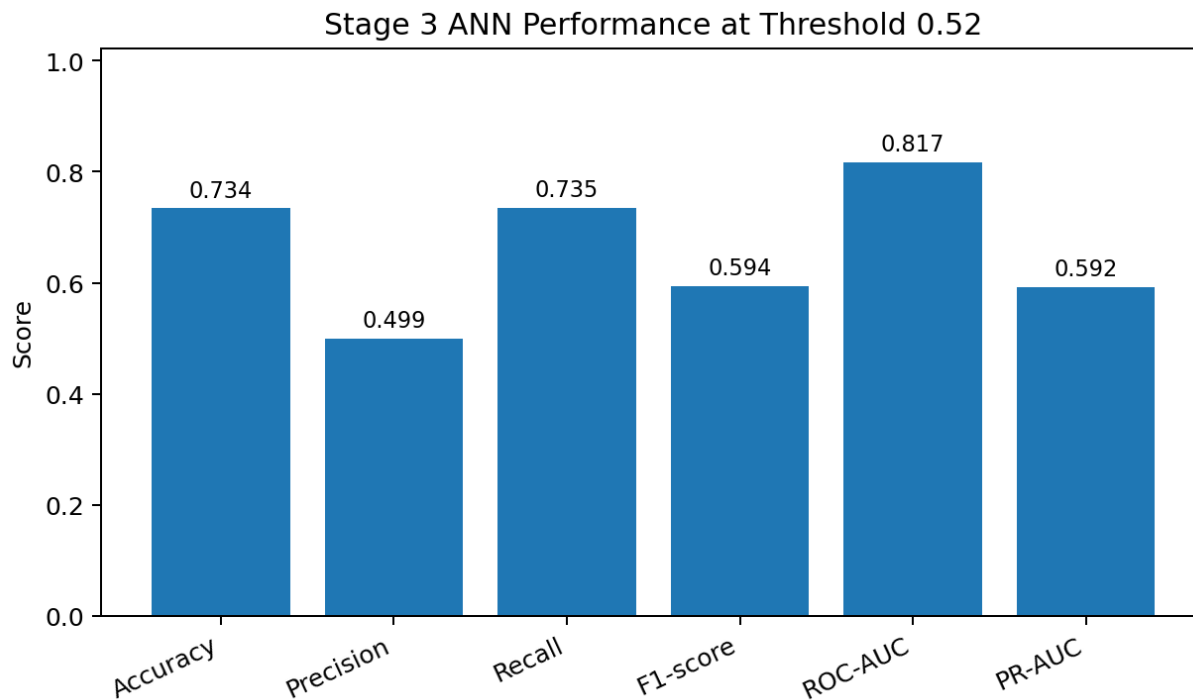


Figure 1 - Stage 3 ANN model performance at threshold 0.52

Because only around 27% of customers churn, accuracy alone is not enough. A simple model that predicted every customer as "no churn" could look accurate but would not help the business retain customers. Therefore, recall, ROC-AUC, PR-AUC, F1-score, and risk ranking should be used when explaining model value.

5. Key Churn Drivers and Feature Importance

The Data Analyst used permutation importance to check which customer attributes the model genuinely relied on. The strongest churn signals were MonthlyCharges and tenure. EstimatedTotalCharges was third, but much weaker than the first two.

Rank	Churn driver	Importance	BA interpretation
1	MonthlyCharges	0.138	By far the strongest churn signal; price pressure and perceived value should be reviewed.
2	tenure	0.119	Almost as strong; customer lifecycle stage is critical.
3	EstimatedTotalCharges	0.013	Relevant for relationship value/history, but much weaker and derived.
4	SeniorCitizen	0.011	Minor effect that may support segmentation, not a standalone campaign driver.
5	IsNewCustomer	0.009	Minor effect but aligns with early-tenure churn risk.

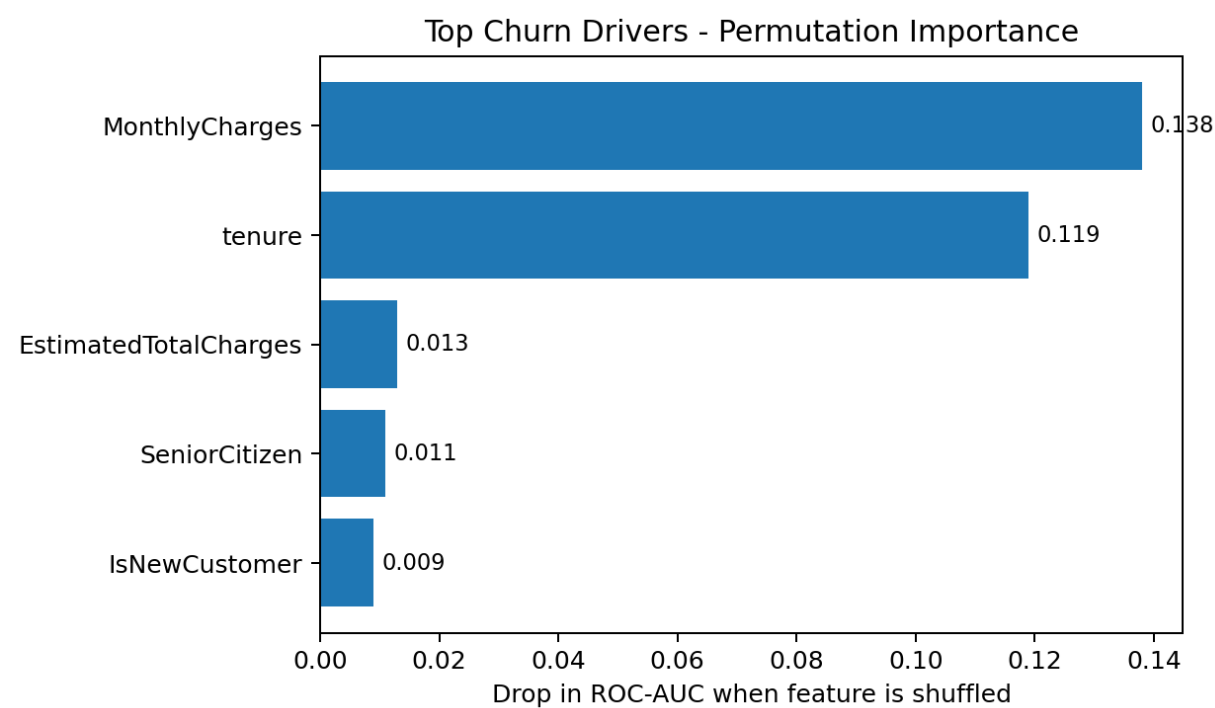


Figure 2 - Top feature importance drivers for churn prediction

BA interpretation: retention strategies should focus on customers with higher monthly charges and shorter tenure. Contract type should not be used as the main campaign driver in this dataset, because the model and earlier statistical testing both showed weak signal for contract type.

6. Tenure-Based Churn Insight

The model results show a clear customer lifecycle pattern: churn is highest during the first year and drops as customers stay longer. This gives the business a clear point for proactive intervention.

Time as customer	Customers	Actual churn rate	BA meaning
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0-12 months	417	45.8%	Highest priority for onboarding and early retention.
13-24 months	176	33.0%	Still high risk; continue proactive engagement.
25-48 months	317	19.6%	Moderate risk; use lighter-touch retention.
49-72 months	439	10.7%	Lowest risk; loyalty management rather than urgent retention.

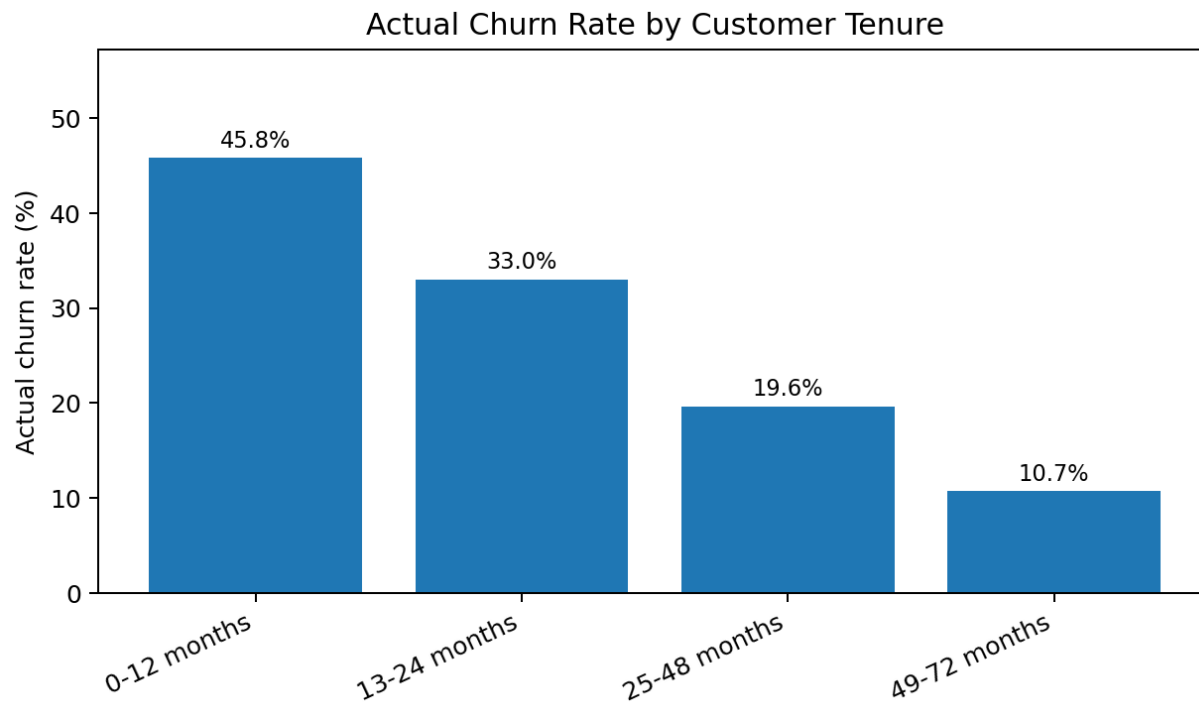


Figure 3 - Churn falls as customer tenure increases

BA interpretation: the first 12 months should be treated as the most important retention window. New customer onboarding, value education, early check-ins, and plan-fit reviews should be implemented before customers become disengaged.

7. Risk Deciles and Top 20% Targeting Insight

The model is valuable because it can rank customers by predicted churn risk. The riskiest 20% of customers contain 44.7% of everyone who actually left. This makes retention activity approximately 2.5 times more efficient than contacting all customers equally.

Risk group	Customers	Actual churn rate	Business meaning
Riskiest 10%	135	69.6%	Highest priority group; urgent retention action is justified.
Next 10%	135	48.9%	Second priority group; still materially above average churn.
Top 20% combined	270	44.7% of all churners	Best campaign starting point for efficient retention spending.
Safest 10%	135	0.0%	Do not spend high-cost retention offers here.

BA interpretation: the business should not use a blanket campaign. Instead, the top risk deciles should receive the first and strongest retention actions, while lower-risk groups should receive lower-cost engagement.

8. Customer Segment Risk

The model outputs were also connected to the team's customer segments. This makes the predictive modelling work easier to explain because customer groups can be described in business language.

Customer segment	Customers	Average tenure	Average monthly bill	Actual churn	BA priority
Short tenure, high charges	419	15.2 months	\$81.98	48.9%	Highest priority: high risk and high value at stake.
Short tenure, low charges	336	11.0 months	\$33.10	23.5%	Medium priority: support onboarding and engagement.
Long tenure, high charges	390	58.4 months	\$91.98	16.7%	Protect loyalty and value; not urgent churn risk.
Long tenure, low charges	204	54.8 months	\$33.63	4.4%	Lowest priority for costly retention campaigns.

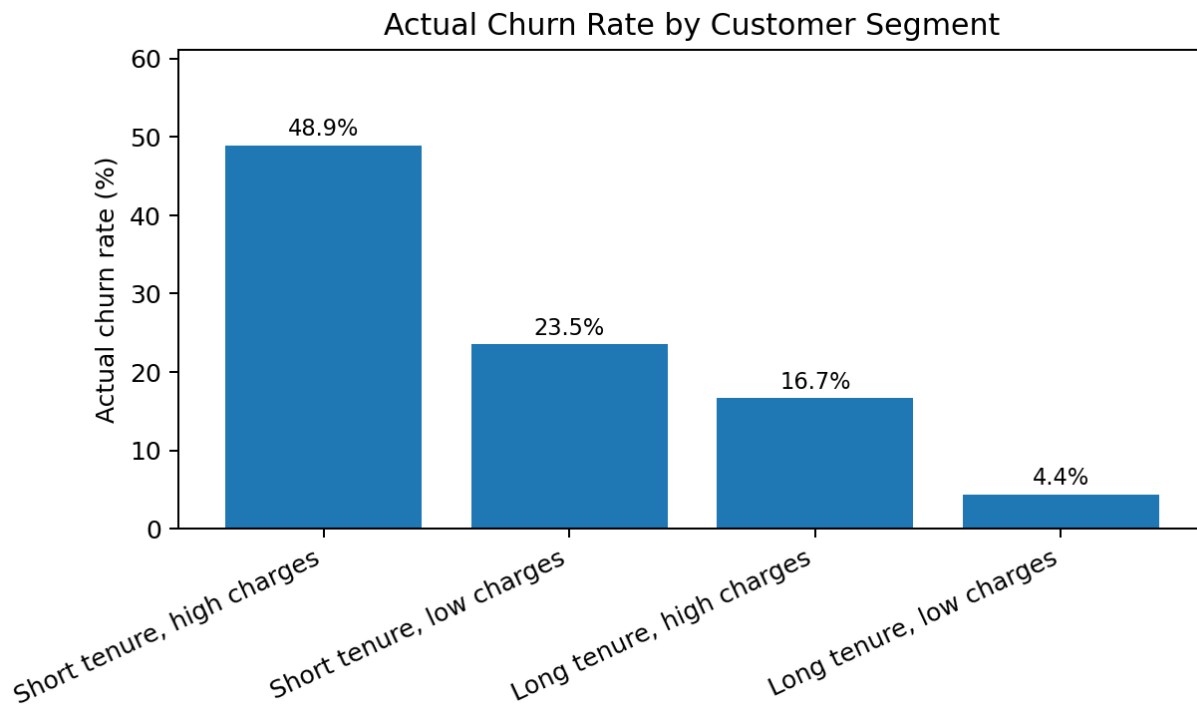


Figure 4 - Churn risk by customer segment

BA interpretation: short-tenure, high-charge customers should be the first priority for retention because they are both more likely to churn and commercially important. The business should review plan fit, billing clarity, customer expectations, and service value for this group.

9. Risk Bands for Retention Action Planning

A list of probabilities is not directly usable by a retention team. Risk bands convert model outputs into practical action groups.

Risk band	Customers	Share of base	Actual churn rate	Share of all leavers	Recommended action
High (>0.70)	284	21.1%	58.8%	46.6%	Direct call, plan review, targeted offer, proactive service support.
Medium (0.40-0.70)	404	29.9%	34.9%	39.4%	Lower-cost digital engagement, education, follow-up campaigns.
Low (<0.40)	661	49.0%	7.6%	14.0%	General loyalty messaging only; avoid expensive incentives.

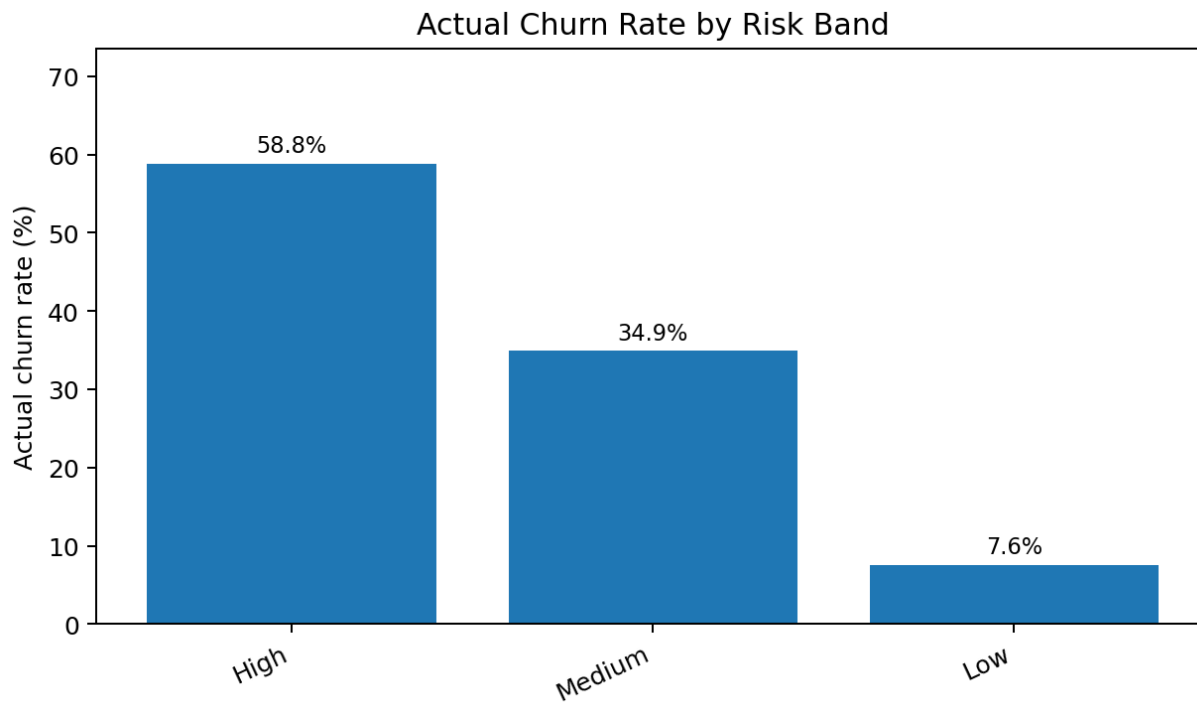


Figure 5 - Actual churn rate by high, medium, and low risk bands

BA interpretation: almost 6 in 10 customers in the high-risk band actually churned, compared with fewer than 1 in 12 in the low-risk band. This shows that the risk bands separate customers well enough to support real retention action planning.

10. Targeted Retention Recommendations

Recommendation	Business reason	Suggested action
Prioritise the top 20% highest-risk customers	They contain 44.7% of actual churners, making retention spending more efficient.	Create a high-risk contact list and action it first.
Focus on the first 12 months	Churn is 45.8% in the first year and falls significantly after longer tenure.	Use onboarding calls, service education, and first-year check-ins.
Review high-charge, short-tenure customers	This segment has 48.9% actual churn and high commercial value at stake.	Offer plan-fit reviews, bill explanation, and personalised retention support.
Use risk bands for action planning	High, medium, and low-risk customers behave very differently.	Match retention cost to risk level.
Avoid contract-only campaigns	Contract type carried weak/no signal in this dataset.	Use churn score, tenure, monthly charges, and risk band instead.
Use the ANN as decision support	Model outputs should support but not replace business judgement.	Combine model scores with complaints, service history, and commercial context.

11. Limitations and Proposed Solutions

Limitation / challenge	Impact on business interpretation	Proposed solution
Missing customer service fields	The data does not include support tickets, complaints, customer satisfaction, payment method, or competitor information, which are usually useful churn predictors.	Collect richer service, complaint, billing, payment, satisfaction, and competitor data in future iterations.
EstimatedTotalCharges is derived	It is calculated from tenure and monthly charges, not actual billed history, so it should not be treated as a real billing record.	Use actual billing and payment history where available.
ANN explainability	Neural networks are harder for non-technical stakeholders to interpret than simpler models.	Use feature importance, customer segments, risk bands, and plain-language explanations.
False positives and false negatives	Some loyal customers may be contacted unnecessarily, and some churners may be missed.	Adjust the decision threshold based on retention budget and business tolerance for missed churners.
Unusual contract-type result	Contract type had weak/no signal even though it is often important in telecom churn.	Validate this finding with the client or a fuller dataset before building contract-based campaigns.
Decision-support only	The model should not make decisions automatically without business context.	Use the model alongside human judgement, customer context, service history, and commercial rules.